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SCHOOL OF PHYSICS
THIRD YEAR LABORATORY

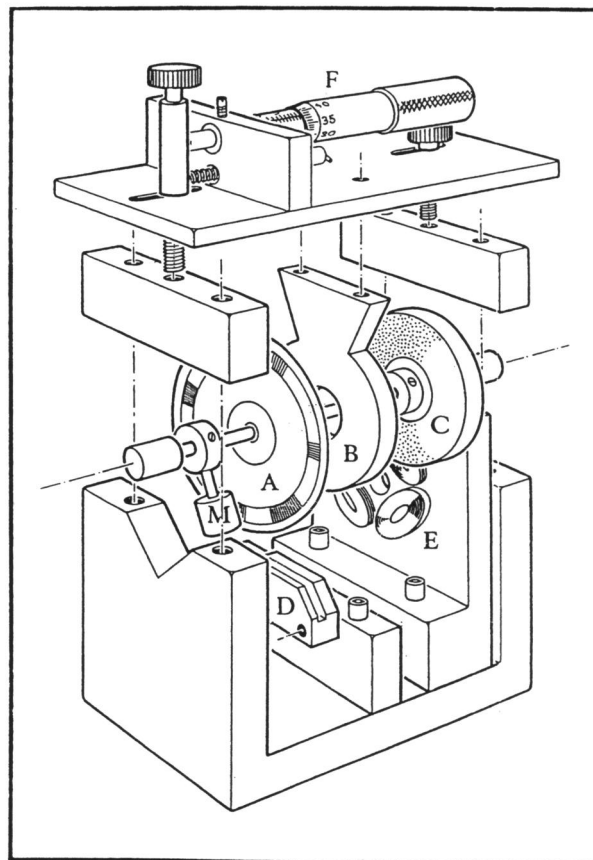
CHAOTIC MOTION

1. INTRODUCTION (Ref. 1, pp 1-5)

The word 'chaos' is used to describe aperiodic, erratic or disordered phenomena in natural or man made systems. Chaos may be **stochastic** (random), in which case there is no discernible pattern (e.g. diffusion, electronic noise) or, as in many systems, it can be **deterministic**, so that a knowledge of the parameters that describe the system, their time-evolution equations and the initial conditions would, classically, be expected to completely determine the behaviour of the system. In a system exhibiting deterministically chaotic behaviour, however, the solutions of the (necessarily) **non-linear** equations describing the system are extremely sensitive to the initial conditions, so that the system's behaviour is non-repetitive but has detectable **patterns**.

In this experiment the study of the motion of a driven, damped pendulum is used to illustrate some of the concepts used in the study of deterministic chaotic phenomena. Even though the pendulum's motion can be completely described by a differential equation, where all the parameters can be defined and measured, the motion becomes chaotic for certain combinations of these parameters. It is our aim, then, to try to find and quantify the order within the apparent disorder.

2. APPARATUS



A Optical Encoder wheel	E Drive Coils
B Copper Damping Plate	F Micrometer
C Annular Ring Magnet	M Pendulum Bob
D Encoder Module	

Fig 1. Exploded schematic diagram of the driven, damped pendulum.

The pendulum consists of an aluminium bob on a rod rigidly connected to a mounting collar near one end of a steel axle, which is supported by ball races in a rigid brass frame. A permanent ring magnet is coaxially mounted near the other end of the axle, with an optical encoder wheel attached coaxially between the bob and the magnet. Torque is applied to the ring magnet by passing a current through drive coils mounted on a printed circuit board next to it. Sinusoidal drive current is supplied by the PENDULUM DRIVER. The drive voltage can be measured using a TEKTRONIX TDS 2002B digital storage oscilloscope. Damping of the pendulum, due to eddy currents induced in a copper damping plate adjacent to the magnet, may be adjusted by varying the magnet - damping plate separation with a micrometer mounted on top of the brass frame.

The optical shaft encoder is used to determine the pendulum's angular position θ , which is recorded by the computer. The θ values are recorded every 2 ms and are then used to calculate the pendulum's angular velocity $d\theta/dt$. Position and velocity data can be saved in an ASCII file, structured as follows: time (sec) in the first column, θ (between π and $-\pi$) in the second column, $d\theta/dt$ in the third column and in the fourth column the number of multiples of 2π which need to be added to the second column to 'unfold' the θ values. For details of the pendulum and its operating software you can consult the "Chaotic Pendulum Manual" on the computer desktop.

3. THEORY

The motion of a sinusoidally driven pendulum of mass m , suspended a distance r from the pivot axis, can be described by the second order differential equation:

$$I d^2\theta/dt^2 + b d\theta/dt + mgr \sin(\theta) = A \cos(\Omega t + \Phi) \quad \dots (1)$$

where:

- θ is the angular position of the pendulum
- I is the total moment of inertia of all the rotating components attached to the axle
- b is the damping coefficient
- A is the amplitude of the applied sinusoidal drive torque
- Ω is the drive frequency of the drive torque
- Φ is the phase of the drive torque

The system's natural frequency is then given by $\omega_0 = \sqrt{(mgr/I)}$.

The nonlinear nature of equation (1) means that solutions exhibiting **sensitive dependence on initial conditions** (i.e. solutions showing chaotic behaviour), are possible for various combinations of the system parameters.

3.1 Phase space and Poincaré sections (Ref. 1, pp 7-13 and pp 21-27)

Phase space is a mathematical space where each of the variables describing the state of a dynamical system is represented by a separate orthogonal co-ordinate direction.

In the case of the pendulum, the motion can be represented in a two dimensional space specified by the angular position and angular velocity of the pendulum.

A **Poincaré** plot (or section) is a stroboscopic or periodic ‘snapshot’ of the variables describing a dynamical system. The sampling period is usually taken as the period of the driving force. Poincaré plots are very useful in simplifying phase space diagrams of complex dynamical systems and are very helpful in describing and characterising chaotic motion.

In the case of our pendulum the Poincaré section is a plot of velocity vs. position, the points on the plot being measured at the same phase in each cycle of the sinusoidal driving torque.

3.2 Some descriptors of chaotic motion

3.2.1 Capacity (or box-counting) dimension (Ref. 1, pp 110–114)

Dimension is a measure of the complexity of the phase space trajectories (or of a Poincaré plot) describing the system. Various dimensions can be defined and calculated. These dimensions are not necessarily integers; they can be fractional and describe the **fractal** geometric objects represented by the trajectories.

In this experiment we calculate the **capacity dimension** by hand. The capacity dimension d , of a set of Poincaré data can be calculated using:

$$d = \ln N(\epsilon) / \ln (1/\epsilon) \quad \dots (2)$$

where ϵ is the box length as a fraction of the total length in that spatial direction, and $N(\epsilon)$ is the number of occupied boxes, i.e. boxes containing at least one Poincaré point. The capacity dimension is thus a measure of how much of the available phase space is occupied by the system’s trajectories.

Problem:

Using equation (2), calculate the capacity dimension in a two dimensional phase space divided into $N \times N$ boxes (where $N = 1/\epsilon$), for the following three special cases:

- a) only one box in the phase space is occupied
- b) a line of boxes, along one of the directions in the phase space, is occupied
- c) every box in the phase space contains at least one point

3.2.2 Lyapunov exponents (Ref. 1, pp 84 – 86 and pp 119 – 121)

The Lyapunov exponent is a measure of the **average** rate at which nearby trajectories in phase space diverge i.e. it is a measure of the rate at which a new trajectory in phase space diverges from the previous trajectory. It is thus a measure of the sensitive dependence on initial conditions that is characteristic of deterministic chaotic behaviour.

Lyapunov exponents give a quantitative measure of a system’s ‘degree of chaoticity’ and can be used to distinguish deterministically chaotic from purely random behaviour. In this experiment the “Chaotic Pendulum” software is used to calculate Lyapunov exponents.

4. EXPERIMENTAL

WARNINGS!

- 1) The pendulum is a **sensitive piece of precision equipment** and needs to be carefully treated at all times. If unsure about a function, **ask** rather than damage!
- 2) The **optical encoder wheel is very fragile**, take care not to touch it or hit it, particularly when 'tapping' the pendulum with a pen/pencil to start it, as in Expt. 1.
- 3) When adjusting the 'damping' micrometer ensure that the **copper damping plate does not foul the ring magnet** or the encoder wheel.

To reset the PENDULUM INTERFACE (when the "Chaotic Pendulum" software seizes up, etc):

- HALT (if necessary) the "Chaotic Pendulum" software and 'Exit program'
- Switch off DRIVE on the PENDULUM DRIVER, pull out the USB cable connected to the socket marked PENDULUM INTERFACE on the computer and wait 10 seconds.
- Make sure the pendulum is **stationary**, plug the USB cable back into the PENDULUM INTERFACE socket and wait until the COMPUTER READY light goes green (the pendulum's position when this light turns green is taken as the 'zero' position).
- Restart the "Chaotic Pendulum" software and switch DRIVE back on. Wait until the motion settles down (1-2 mins).

Note: When running the "Chaotic Pendulum" software, all other applications need to be closed (or at least minimised)

Expt. 1: Damped harmonic motion and phase space trajectories

Make sure the PENDULUM DRIVER power is switched off and the USB cable from the PENDULUM INTERFACE is not plugged into the computer. Set the damping micrometer to ~16 mm

Gently tap the pendulum with the blunt end of a pencil so that it oscillates; measure the pendulum's period using a stop watch and hence calculate the pendulum's **natural frequency**.

When the pendulum is **completely at rest** again, plug the USB cable from the INTERFACE into the computer (into the USB socket labelled PENDULUM INTERFACE), wait until the 'COMPUTER READY' light on the INTERFACE goes green and start the "Chaotic Pendulum" application.

Tap the pendulum hard enough (but **not too violently!**) so that it rotates, press the 'Plot data' button (the HALT button will turn green) then press 'Clear' just when the pendulum stops rotating and starts oscillating. When the pendulum comes to rest press HALT and 'Print' the screen. This plot (velocity vs phase) shows the trajectory of the lightly damped pendulum motion in the phase space (which, in this experiment, is a plane).

Now press 'Save orbit', enter 10,000 as the 'number of data points' and 'Create New Folder' (with your surname), in the 'My Documents / Daedalon' folder, to store your experimental data. Double click your data folder's name to open it.

To save the pendulum trajectory data, enter a 'File name' but do NOT press 'Save' yet...

Now tap the pendulum with the pencil (enough to make it rotate full revolutions) and press 'Save' just as the pendulum stops rotating and starts oscillating. The HALT button turns red while data is being recorded and saved.

Orbit data is saved in ASCII format in the PHA file. The first line gives the number of data points stored in the file, the second line specifies the data type (PHASEE); the numerical data is then formatted one data point per line, in four columns, separated by spaces. The first column is the time of the reading (in seconds), the second is the measured position coordinate (angle, in radians) the third is the calculated angular velocity (rad/s) and the fourth is the number of times the pendulum has rolled over since the beginning of data recording in the file.

When data recording is finished (HALT goes grey again) 'Print' the screen and press 'Exit program'. Following instructions in the information folder, use "Excel" to produce a plot of the pendulum's position vs. time.

Now set the damping micrometer to 13.0 mm and repeat the relevant procedures above to obtain:

- a) a printout of the pendulum's trajectory (i.e. velocity vs position) in phase space
- b) a plot of pendulum position vs. time with 13 mm damping.

Compare a) and b) with the corresponding plots for 16 mm damping and comment on the similarities and differences.

Question 1: *What would the trajectory of an ideal **undamped** pendulum look like?*

Open the '13 mm' orbit data file with "Excel". Find the times (first 20, say) at which the velocity is zero and note the **magnitude** of the angular position (= amplitude) at each of these times. From a plot of the time difference between consecutive zeros versus the corresponding average amplitude, estimate the pendulum's natural period (= period when the amplitude $\rightarrow 0$). Compare with the pendulum period measured using the stopwatch.

Expt. 2: Single and bifurcated orbits - period doubling

Set the micrometer to 8.0 mm, turn the AMPLITUDE knob on the PENDULUM DRIVER fully anticlockwise (minimum) and switch the POWER and the DRIVE on.

Reset the PENDULUM INTERFACE, as described in the box above..

Use the CH1 VERTICAL input of the “TEKTRONIX” CRO (input DC coupled!) to MEASURE the peak-to-peak (CH1 Pk-Pk) drive voltage out of the sockets labelled DRIVE AMPLITUDE at the back of the PENDULUM DRIVER. Set the FREQUENCY about 30% above the pendulum’s natural frequency (i.e. to ~ 1.80 Hz - in order to get chaotic motion, the pendulum usually needs to be driven at a frequency some 30% different to its **natural frequency**)

Slowly increase the drive voltage (AMPLITUDE), stopping frequently to observe the pendulum’s trajectory. When the Pk-Pk voltage is ≥ 1.5 V and you have a **single** ‘orbit’ (approx elliptical) on the screen, let the system settle for ~ 1 min, note the Pk-Pk drive voltage then press ‘Plot data’. Record data for about 1 minute then press HALT and ‘Print’ the screen.

Explain the features of the printed orbit and of the Poincaré plot.

Now ‘Save orbit’ (10,000 points) in your data folder. When data recording is finished, use “Excel” to print a graph of position vs. time for the single orbit.

Explain the features of your position vs. time plot.

Slowly increase the drive voltage again until the orbit bifurcates, i.e. until you get a well separated **double** orbit. Let the system settle for ~ 1 min, note the Pk-Pk drive voltage and again ‘Plot data’ for about 1 minute then HALT and ‘Print’ the screen.

Compare with the single orbit screen printout above and comment on the differences (particularly in the Poincaré plot).

Save the double orbit (10,000 pts) and graph (“Excel”) position vs. time. Compare with the corresponding graph for the single orbit. Comment on the similarities and differences.

Question 2: What does ‘period doubling’ refer to?

Expt. 3: Chaotic motion

Note: In this part of the experiment try to save all the chaotic motion data files in the course of the **same experimental session** so that the AMPLITUDE and DRIVE FREQUENCY settings remain **exactly** the same.

With the damping micrometer set to 8.0 mm slowly increase the driving voltage (beyond that needed for the double orbit of Expt. 2) until the orbits become non-repetitive. (To check this, ‘Clear’ the screen each time you change the AMPLITUDE)

When about to ‘Save’ data (orbit or Poincaré), first allow the motion to stabilise for ~ 1 min. If the pendulum settles into regular motion, slightly increase the AMPLITUDE and again leave to stabilise ~ 1 min before starting to ‘Save’ data.

If the “Chaotic Pendulum” software stops during a ‘Save’, you will need to reset the PENDULUM INTERFACE and start the data acquisition again.

Save the following data files (remember to let the pendulum settle ~ 1 min after any change in pendulum parameters), keeping the drive voltage and frequency exactly the same:

- a) 8.0 mm damping: 37,000 point orbit data file (~ 80 sec)
- b) 8.0 mm damping: reset the PENDULUM INTERFACE, then save a 500 point Poincaré data file (~ 4.5 mins)
- c) set the damping to 9.0 mm and save a 37,000 point orbit data file (~ 80 sec)
- d) 9.0 mm damping: reset the PENDULUM INTERFACE and save a 500 point Poincaré data file (~ 4.5 mins)

3.1 Capacity (or box-counting) dimension

Poincaré data is saved in ASCII format as follows: the first line of the POI data file gives the number of data points stored in the file, the second line specifies the data type (POINCAREE); the numerical data is then formatted one data point per line, in two columns, separated by spaces. The first column is the measured position coordinate (angle, in radians) and the second is the calculated angular velocity (rad/s).

Following instructions in the information folder use “Excel” to plot and print the Poincaré section of the 8 mm damping data obtained in Expt. 3 b).

In the plot use:

X Scale:- Minimum: -4	Maximum: 4	Major unit: 2
Y Scale:- Minimum: -24	Maximum: 8	Major unit: 8

This should give a printout with 4 major division gridlines in both the X and the Y directions.

Repeat the plot with X and Y Scale major units of (1, 4), (0.5, 2) and (0.25, 1) to give printed plots with $(1/\epsilon =)$ 8, 16 and 32 divisions, respectively, on each axis.

For each of the above printed plots count the number of boxes N , which is occupied and plot $\ln N$ vs. $\ln (1/\epsilon)$. The gradient of this last plot is then the capacity dimension (see equation (2)) for the driven pendulum motion with 8.0 mm damping.

Now plot (32 divisions on each axis) and print the Poincaré section of the 9.0 mm damping data obtained in Expt. 3 d). Count the number of occupied boxes and, using eqn (2), calculate the capacity dimension. How does this compare with the capacity dimension for 8.0 mm damping found above? Is this the trend you would (naively) expect? Explain why.

3.2 Lyapunov exponents (Ref. 1, pp 120 – 121)

When 'FFT' is pressed and an 'orbit' (PHA) data file is nominated, the "Chaotic Pendulum" software first calculates the Fast Fourier Transform (or power spectrum of the pendulum system; see Ref. 1, pp 27 – 30), using the velocity data in the file; it then calculates the Lyapunov exponent, i.e. the average rate of divergence of neighbouring trajectories.

Calculate and 'Print' the FFT power spectra (using the 'Welch' window function), and note the Lyapunov exponents for the 8.0 mm and 9.0 mm damping orbit data files obtained in Expts. 3a) and 3c) above.

Note at which frequencies peaks occur in the FFT spectra. How are these peak frequencies related to frequencies characteristic of your pendulum system?

Compare the calculated Lyapunov exponents for the 8.0 mm and 9.0 mm damping. What would you, naively (backed up by the capacity dimension data for 8.0 mm and 9.0 mm damping, from Expt. 3.1) expect the trend to be as the damping decreases? Explain.

Expt. 4: Simulated chaotic motion of the pendulum

Pressing the 'Simulation' button activates a software routine which calculates the numerical solution of the equation of motion of a damped driven pendulum (eqn (1)).

The units of the pendulum parameters are correct as given on the screen, apart from the unit for moment of inertia (I) which should be kg m^2 . Poincaré data is recorded when modulo $\Omega t = 2\pi$

The simulation software starts up with a default set of pendulum parameters and the simulation can be carried out either in 'Real time' or, much faster, in 'Computer time'. Simulated data can be saved to ASCII files ('Phase plot' or 'Poincaré plot') formatted in the same way as the experimental data files acquired with 'Save orbit' and 'Save poincare' respectively.

(Time out allows you to set a 'settling' period, where data is calculated but is not displayed or recorded.)

4.1 Capacity dimension vs. number of data points in a Poincaré plot

With the simulation parameters set at their default values generate ('Computer time') and save Poincaré files with 100, 200, 400, 800, 1200, 1600 and 2000 points.

Using the 'Dimension' button find the capacity dimension for each data set and plot dimension vs number of data points.

Explain (qualitatively) the behaviour of your plot. What is the least number of points needed for a consistent estimate of dimension to be made?

4.2 The 3-dimensional nature of the pendulum's Poincaré sections

In this section you will show that Poincaré plots of the pendulum, displayed on the screen, are slices from a three dimensional plot of pendulum position, velocity and the **drive phase** of the pendulum driver.

With all the other simulation parameters set at their default values, generate ('Computer time') and save Poincaré files (each with 1000 points) with 'Drive phase' of 0.0, 0.4, 0.8, 1.2, 1.6, 2.0, 2.4, 2.8, and 3.2 radians.

To display and compare Poincaré sections with different phases, they will need to be printed and then photocopied onto transparencies.

Follow the instructions in the information folder and use "Excel" to print the Poincaré sections for each of the nine 'Drive phase' settings. In all the 9 plots **untick** the Major gridlines for both the (X) and the (Y) axis and use exactly the same axis scaling i.e:

X Scale :- Minimum: -4 Maximum: 4
Y Scale :- Minimum: -25 Maximum: 25

Set a photocopier to **reduce to half the original area** (i.e. A3 $\rightarrow$ A4) and photocopy the printed Poincaré sections, pairwise, onto **A4 transparencies**. Carefully cut each transparency into halves and mount the cut transparency halves in order of increasing phase, on the CD rack supplied.

Viewing all the mounted transparencies from above, at about 45°, will reveal the 3 dimensional character of the full Poincaré plot

Question 3: Find the capacity dimension (using the 'Dimension' button) of any three of the 9 simulated Poincaré data sets above. Do the 3 values differ significantly? Qualitatively, explain why.

5. REFERENCES

1. Baker G. L. And Gollub J. P., "Chaotic dynamics: an introduction" 2nd ed. (1996)
2. Hilborn R. C., "Chaos and nonlinear dynamics" (1994)